

Caoliwen Wang

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Research Interests

Rooted in vision, robotics, generative and agentic AI, my research develops representations of world modeling across **language, vision and physics** modalities, enabling AI to perceive, model, and generate physical world.

Education

University of British Columbia

Sep 2025 - Present

Ph.D. in Computer Science

Advisor: Prof. Peter Yichen Chen

University of Science and Technology of China

Sep 2021 - Jun 2025

B.S. in Computational Mathematics

Advisor: Prof. Ligang Liu

Research Experience

Language for World Modeling

Feb 2026 - Present

- **WorldAgent:** Developed a verification-guided agentic pipeline that leverages language and 3D modalities to construct editable and controllable physical worlds.
- **Agentic Real-to-Sim:** Developed an agentic pipeline that leverages vision to automatically bridge the real-to-sim gap for robotic manipulation of diverse powders, generating scalable training data for policies that generalize across powder materials.

Physics for World Modeling

Jun 2025 - Present

- **WorldParticle:** Developed a unified model that represents diverse physical phenomena as Lagrangian particles and learns their dynamics with a Transformer-based backbone, enabling fully data-driven simulation without the hand-crafted design and pre-defined physical biases required by traditional solvers, toward unified and scalable world modeling.
- **WorldParticle++:** Built upon the WorldParticle backbone, scaling particle-based simulation to million-scale systems with support for dynamic boundaries, while leveraging forcing techniques from video-based diffusion models to improve rollout stability for long-horizon simulation.
- **Physically Plausible 3D Configuration Generation:** Developed a method that enforces physical validity, including non-penetration and structural plausibility, on generated 3D protein configurations, while improving the inference efficiency of protein structure prediction foundation models.

Vision for World Modeling

Jun 2026 - Present

- **Force-conditioned Streaming Video Generation:** Developed a force-conditioned streaming video world model pipeline enabling perception of diverse materials and supporting diverse physical interactions.

Design-Driven 3D Shape Generation

Jan 2024 - Jun 2025

- **Neural Shadow Art:** Developed an implicit neural 3D representation that facilitates geometry generation from multi-view binary shadow targets, with angular adjustment and volume/geometry optimization.
- **Shape from Semantics:** Developed a 3D generation framework for creating semantically consistent shapes whose multi-view renderings align with user-specified prompts, extending the inverse-modeling framework to a semantics-conditioned 3D generation problem.

Industry Experience

Style3D

Jun 2026 - Present

- **WorldContact:** Developed a unified world simulator that handles diverse contact interactions and efficiently generates high-quality data for scalable robot learning, demonstrating its potential to close the real-to-sim-to-real loop and enable Physical Recursive Self-Improvement.

Mentor: Prof. Huamin Wang

Publications & Preprints

- [1] **Caoliwen Wang**^{*}, Mengdi Wang^{*†}, Yige Chen^{*}, Zejia Wu, Bowen Huang, Siyuan Chen, Guanxiong Chen, Lifu Wei, Heng Zhang, Qinghai Zhang, Yin Yang, Guandao Yang, Shiyong Xiong, Peng Wang, Chenfanfu Jiang, and Peter Yichen Chen. **WorldAgent: Verification-Guided Agentic Physical World Construction**. *Under Review*.
- [2] **Caoliwen Wang**, Mengdi Wang[†], Heng Zhang, Shixun Huang, Siyuan Chen, Chao Liu, Anpei Chen, Zhendong Wang, Peter Yichen Chen[†], and Huamin Wang. **WorldContact: A Contact-Centric World Model for Scalable Robot Learning**. *Under Review*.
- [3] **Caoliwen Wang**^{*}, Minghao Guo^{*†}, Siyuan Chen^{*}, Heng Zhang, Mengdi Wang, Xingyu Ni, Hanson Sun, Kunyi Wang, Zherong Pan, Kui Wu, Lingjie Liu, Yin Yang, Chenfanfu Jiang, Taku Komura, Wojciech Matusik, and Peter Yichen Chen[†]. **WorldParticle: Unified World Simulation of Lagrangian Particle Dynamics via Transformer**. *SIGGRAPH Asia (Conference Track)*, 2026.
- [4] Xingyu Ni[†], Jiong Chen, Siyuan Chen, **Caoliwen Wang**, Mathieu Desbrun, and Taku Komura. **Efficient Open-Boundary Poisson Solves**. *ACM Transactions on Graphics (Proc. SIGGRAPH Asia)*, 2026.
- [5] Siyuan Chen^{*}, Minghao Guo^{*†}, **Caoliwen Wang**, Anka He Chen, Yikun Zhang, Jingjing Chai, Yin Yang, Wojciech Matusik, and Peter Yichen Chen[†]. **Physically Valid Biomolecular Interaction Modeling with Gauss-Seidel Projection**. *ICLR*, 2026.
- [6] Liangchen Li, **Caoliwen Wang**, Yuqi Zhou, Bailin Deng, and Juyong Zhang[†]. **Shape from Semantics: 3D Shape Generation from Multi-View Semantics**. *Under Review*.
- [7] **Caoliwen Wang**, Bailin Deng, and Juyong Zhang[†]. **Neural Shadow Art**. *Pacific Graphics (Conference Track)*, 2025.

Awards

University and Provincial Outstanding Graduate Award 2025 (USTC)

Volunteering

Conference Volunteer

Chinagraph, 2024 | *Huangshan, Anhui, China*
 Pacific Graphics, 2024 | *Huangshan, Anhui, China*

Oct 2024
 Oct 2024